CPTP-Compiler-PINNs: Structure-Preserving Residual Learning and Matrix-Free Compilation for Open Quantum Dynamics

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Abstract

Learning open quantum dynamics from sparse, noisy observations is a central task in quantum device characterization, noise modeling, and control. Standard neural surrogates for density matrices can leave the physical state space during training, and dense Liouvillian implementations materialize $d^2 \times d^2$ superoperators even when the physical model is specified by only a few local Hamiltonian and dissipative terms. We introduce *CPTP-Compiler-PINNs*, a structure-preserving residual-learning framework for Markovian open quantum systems. The method combines a Cholesky density network that maps every time point to a positive semidefinite, unit-trace density matrix; a softplus-positive Lindblad-rate (or positive Kossakowski) parameterization that preserves the Gorini–Kossakowski–Sudarshan–Lindblad generator form; and a matrix-free compiler that lowers a symbolic open-system specification to either dense or structured residual kernels. We prove exact physicality of the density map, universal approximation of continuous density trajectories, complete-positivity and trace preservation of the learned generator, a trace-norm residual certificate for trajectory error, a local identifiability theorem, and a dense-versus-structured compiler complexity theorem. Reproducible experiments support the central claims on a single-qubit inverse problem and a sparse-measurement ablation—where the constrained model recovers Hamiltonian and dissipative parameters, attains near-unit trajectory fidelity, outperforms an unconstrained baseline, and remains accurate as observations are removed—and on a dense-versus-structured compiler benchmark. On a two-qubit dissipative gate the recovered generator generalizes to held-out initial states at ≈ 0.98 fidelity, and a qutrit study shows that Fourier time-features overcome the spectral bias of a plain temporal network, raising reconstruction fidelity from 0.88 to 0.95 (precise rate identification in that fast regime is left to future work). The simulations are intentionally reproducible and modest in scale; they are not hardware data or state-of-the-art claims. The contribution is a mathematically constrained, accelerator-aware formulation that scales with batched GPU kernels and integrates with quantum software stacks.

1 Introduction

Open quantum systems are the operational language of modern quantum hardware. A superconducting, trapped-ion, photonic, spin, or neutral-atom processor is never described only by its ideal Hamiltonian; it is also described by relaxation, dephasing, leakage, crosstalk, drift, and control-induced dissipation. The mathematical object that must remain physical throughout this description is the density matrix. For a finite Hilbert space $\mathcal{H} \cong \mathbb{C}^d$, the state space is

$$\mathcal{D}(\mathcal{H}) = \{\rho \in \mathbb{C}^{d\times d} : \rho = \rho^\dagger,\ \rho \succeq 0,\ \operatorname{Tr}\rho = 1\}. \tag{1}$$

When the dynamics are Markovian and time-local, the most general generator of a completely positive trace-preserving (CPTP) semigroup has the Gorini-Kossakowski-Sudarshan-Lindblad (GKSL) form [1–3]. This form is the backbone of quantum noise modeling, quantum control, and calibration workflows [4–7].

The inverse problem is increasingly important: infer a physically admissible dynamical model from incomplete observations. Classical process tomography reconstructs a channel but scales poorly with system size [8–10]. Hamiltonian learning and Lindblad learning exploit dynamical structure but still face identifiability and optimization challenges [11, 12]. Neural methods and physics-informed neural networks (PINNs) offer flexible function approximation with equation residuals in the training objective [13–15]. Yet a direct neural map $t \mapsto \rho_\theta(t)$ usually has no reason to remain Hermitian, positive semidefinite, or unit trace. A loss penalty for negative eigenvalues is not equivalent to physicality during training; it merely asks the optimizer to repair an invalid output after the fact.

This paper takes the opposite design principle: *make invalid states unrepresentable.* The proposed density network is

$$\rho_\phi(t) = \frac{A_\phi(t)A_\phi(t)^\dagger}{\mathrm{Tr}[A_\phi(t)A_\phi(t)^\dagger]}, \tag{2}$$

where $A_\phi(t)$ is a neural lower-triangular complex matrix with nonzero diagonal. This Cholesky-type map is differentiable, architecture-level physical, and compatible with either automatic differentiation or stable finite-difference residuals. Dissipation rates are parameterized by softplus transforms, and full Kossakowski matrices can be represented as $BB^\dagger$. The learning objective combines data likelihood, initial-state matching, and a Lindblad residual evaluated by a compiler that can avoid explicit $d^2 \times d^2$ Liouvillians.

The word "compiler" is used deliberately. A quantum model is specified in an intermediate representation (IR): Hilbert dimension, Hamiltonian terms, jump operators, rates, observables, batching rules, and differentiation mode. The compiler lowers this IR to either a dense Liouvillian or a structured residual kernel. For small d, dense kernels can be convenient. For larger d, the structured route stores H and m Lindblad operators, not a $d^2 \times d^2$ matrix. This aligns with the design philosophy of contemporary quantum and accelerator software: represent intent at a high level, then lower to hardware-conscious kernels. In that sense, the framework is naturally compatible with PyTorch [16], SciPy [17], open-system toolkits such as QuTiP [18], circuit frameworks such as Qiskit [19], GPU-accelerated quantum simulation backends, and future custom GPU kernels.

1.1 Contributions

The core contributions are:

1. **A hard-constrained density network.** The Cholesky density map enforces Hermiticity, positive semidefiniteness, and unit trace exactly at every time point, every training iteration, and every evaluation point.

2. **A CPTP generator parameterization.** Nonnegative Lindblad rates are represented by softplus parameters; full Kossakowski matrices are represented as positive semidefinite Gram matrices. The learned generator therefore remains in GKSL form.

3. **A matrix-free open-system compiler.** A symbolic open-system IR is lowered to either a dense superoperator or a structured residual kernel. The theorem in Section 5.6 quantifies memory and arithmetic cost.

4. **Six mathematical guarantees.** We prove physicality, universal trajectory approximation, CPTP semigroup preservation, a residual-to-error certificate in trace norm, a local identifiability bound, and compiler complexity.

5. **A reproducible numerical study.** All figures and tables are generated by the accompanying scripts. A single-qubit inverse problem and a sparse-measurement ablation show accurate parameter and trajectory recovery that improves on an unconstrained baseline; a compiler benchmark confirms the structured-versus-dense cost advantage; a two-qubit dissipative gate shows that the recovered generator generalizes to held-out initial states (≈ 0.98 fidelity); and a qutrit study shows that Fourier time-features overcome spectral bias (reconstruction fidelity $0.88 \rightarrow 0.95$). Precise rate identification in the fast-oscillatory regime is the main open problem.

The results are intentionally stated conservatively. They show the value of the structural constraints and the compiler path on controlled simulations. They do not claim quantum-hardware validation, universal superiority over all baselines, or production-scale GPU performance. Those are future experimental targets, not present facts.

2 Related work and positioning

2.1 Open-system identification and tomography

The traditional route to learning quantum dynamics is tomography. Quantum process tomography estimates a channel from tomographically complete preparations and measurements [8, 9]. It is model agnostic, but the number of settings grows rapidly with Hilbert-space dimension. Completely positive projection methods can repair an estimated map, but a projection step is not the same as learning inside the physical set throughout optimization [10]. Lindblad identification methods exploit the generator structure and can reduce the number of parameters when the dissipative mechanisms are known or partially known [11]. Hamiltonian learning similarly exploits a structured dynamical model and has been demonstrated experimentally [12]. The present method follows the structured-identification philosophy but replaces unconstrained trajectory estimation by a hard density-matrix parameterization.

2.2 Physics-informed machine learning

PINNs were introduced as a general framework for solving forward and inverse problems by penalizing differential-equation residuals at collocation points [13]. Their success depends on optimization, sampling, scaling of loss terms, and causal or curriculum training strategies [14, 20]. The neural tangent kernel literature helps explain some optimization behavior in overparameterized networks [21], while neural ODEs provide a complementary viewpoint in which the differential equation itself is a learnable continuous-depth model [15]. Operator-learning models such as DeepONet and Fourier neural operators target families of solution maps rather than a single trajectory [22, 23]. CPTP-Compiler-PINNs are closest to inverse PINNs: the network represents a trajectory, and generator parameters are inferred by matching both data and physics residuals.

2.3 Quantum neural representations

Neural quantum states have become a powerful representation for many-body wavefunctions and density operators [24, 25]. Variational neural ansatzes have also been applied to open-system steady states [26]. Quantum machine learning more broadly studies quantum feature spaces, hybrid algorithms, and variational circuits [27, 28]. These works are complementary. The present manuscript does not propose a quantum circuit ansatz or a many-body wavefunction ansatz. It proposes a constrained classical surrogate for density-matrix trajectories and a compiler path for evaluating Lindblad residuals efficiently.

2.4 Dissipation as a resource and non-Markovian extensions

Dissipation is not only a nuisance; it can be engineered for state preparation and quantum computation [29]. Open quantum walks and non-Markovian models show that richer dynamical structures may be necessary outside the GKSL semigroup setting [30–33]. The method here deliberately starts with the Markovian case because it is the right abstraction for many calibration and noise-modeling tasks. The same density parameterization could be used inside a non-Markovian embedding, but the generator theorem would need to be replaced by a theorem for the enlarged model.

3 Background

3.1 GKSL dynamics

Let $\rho(t) \in \mathcal{D}(\mathcal{H})$ be a density matrix. A time-independent Markovian master equation has the form

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = \mathcal{L}_\Theta[\rho] = -\mathrm{i}[H_\Theta, \rho] + \sum_{k=1}^{m} \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\} \right), \tag{3}$$

where $H_\Theta = H_\Theta^\dagger$, $\gamma_k \geq 0$, and the L_k are jump operators. The anticommutator is $\{A, B\} = AB + BA$. A time-dependent generator $\mathcal{L}_t$ can also be used provided it remains in GKSL form at each time. The original GKSL theorem states that this structure characterizes generators of quantum dynamical semigroups in finite dimensions [1, 2]. Standard treatments of open quantum systems and quantum measurement build on this fact [3, 4].

For vectorization, we use column-major convention $\mathrm{vec}(A\rho B) = (B^T \otimes A)\,\mathrm{vec}(\rho)$. The dense Liouvillian associated with (3) is

$$\begin{aligned} \mathbb{L}_\Theta &= -\mathrm{i}(I \otimes H_\Theta - H_\Theta^T \otimes I) && (4) \\ &\quad + \sum_k \gamma_k \left(L_k^* \otimes L_k - \frac{1}{2} I \otimes L_k^\dagger L_k - \frac{1}{2}(L_k^\dagger L_k)^T \otimes I \right). && (5) \end{aligned}$$

This representation is useful for small systems, but it stores d^4 complex scalars. In contrast, direct evaluation of (3) stores H and the L_k matrices, requiring $O(md^2)$ storage.

3.2 Physics-informed learning

A PINN approximates a trajectory or field by a neural network and penalizes the governing-equation residual [13, 14]. For an open quantum system, a generic residual objective is

$$\mathcal{J}(\phi, \Theta) = \lambda_{\mathrm{data}} \mathcal{J}_{\mathrm{data}} + \lambda_{\mathrm{phys}} \mathcal{J}_{\mathrm{phys}} + \lambda_0 \mathcal{J}_0. \tag{6}$$

The data term compares predicted observables with measurements, the physics term penalizes the master-equation residual, and the initial-condition term anchors $\rho(0)$. Neural ODEs, DeepONets, and Fourier neural operators provide related operator-learning approaches [15, 22, 23]. Hamiltonian, Lagrangian, and other structure-preserving networks show that hard physical constraints can improve scientific learning by restricting the hypothesis class [34–36].

Quantum machine-learning work has demonstrated neural state representations, quantum-state tomography, variational steady-state ansatzes, and non-Markovian dynamics models [24–26, 32, 33]. The present work is narrower: it focuses on Markovian density-matrix trajectories and asks that both the trajectory surrogate and the generator parameterization remain physically valid by construction.

3.3 Why soft penalties are not enough

Suppose a neural network outputs an arbitrary Hermitian trace-one matrix. The Bloch-sphere analogy shows the issue immediately: a qubit matrix

$$\rho(r) = \frac{1}{2}(I + r_x\sigma_x + r_y\sigma_y + r_z\sigma_z) \tag{7}$$

is a density matrix if and only if $\|r\|_2 \leq 1$. A neural network with unconstrained $r(t) \in \mathbb{R}^3$ can leave the ball even when the loss eventually pushes it back. During optimization, such nonphysical values can corrupt residuals, produce negative probabilities, and interact badly with parameter estimation. Projection after each forward pass solves physicality only after a discontinuous or non-smooth operation. The Cholesky map instead defines the network image inside $\mathcal{D}(\mathcal{H})$.

4 Method: CPTP-Compiler-PINNs

4.1 Hard-constrained density parameterization

Let $B_\phi : [0,T] \to \mathbb{C}^{d\times d}$ be a neural network whose output is interpreted as a lower-triangular matrix. We use a nonzero diagonal to avoid a zero denominator:

$$A_\phi(t)_{ij} = 0 \quad (i<j), \qquad A_\phi(t)_{ii} = a_{\min} + \mathrm{softplus}(b_{ii}(t)), \qquad A_\phi(t)_{ij} = b_{ij}(t) \quad (i>j), \tag{8}$$

where $a_{\min} > 0$ is a small numerical floor. The density output is

$$\rho_\phi(t) = \frac{A_\phi(t)A_\phi(t)^\dagger}{\mathrm{Tr}(A_\phi(t)A_\phi(t)^\dagger)}. \tag{9}$$

The lower-triangular structure is not essential for physicality, but it removes redundant degrees of freedom and connects the representation to Cholesky factorization.

The output dimension is d^2 real degrees for a complex lower-triangular matrix with real positive diagonal: d real diagonal entries and $d(d-1)$ real off-diagonal degrees. A full-rank density matrix has $d^2 - 1$ real degrees after trace normalization, so the representation is only mildly redundant. Rank-deficient states are obtained as limits when some diagonal entries approach the floor and the off-diagonal factors align accordingly.

4.2 Generator parameterization

For known jump operators with unknown rates, we write

$$\gamma_k(\xi_k) = \mathrm{softplus}(\xi_k) + \gamma_{\min}, \qquad \gamma_{\min} \geq 0. \tag{10}$$

For a full Kossakowski matrix in a fixed operator basis $\{F_a\}_{a=1}^q$, we use

$$C_\eta = B_\eta B_\eta^\dagger \succeq 0, \tag{11}$$

and write

$$\mathcal{L}_\Theta[\rho] = -\mathrm{i}[H_\Theta, \rho] + \sum_{a,b=1}^{q} (C_\eta)_{ab} \left(F_a \rho F_b^\dagger - \frac{1}{2}\{F_b^\dagger F_a, \rho\} \right). \tag{12}$$

Hamiltonians are parameterized in a Hermitian basis $\{G_\ell\}$:

$$H_\theta(t) = \sum_\ell \theta_\ell(t) G_\ell, \qquad G_\ell = G_\ell^\dagger. \tag{13}$$

Together, these choices ensure that the learned generator remains in GKSL form.

4.3 Observation model and loss

Let $\{O_j\}_{j=1}^{p}$ be measured observables, let y_{ij} be a measurement at time t_i, and let $m_{ij} \in \{0,1\}$ indicate whether the observation is present. The data loss is

$$\mathcal{J}_{\text{data}}(\phi) = \frac{1}{\sum_{i,j} m_{ij}} \sum_{i,j} m_{ij} \left(\operatorname{Tr}[O_j \rho_\phi(t_i)] - y_{ij}\right)^2 . \tag{14}$$

The physics loss is

$$\mathcal{J}_{\text{phys}}(\phi, \Theta) = \frac{1}{N_c} \sum_{\ell=1}^{N_c} \|\dot{\rho_\phi}(\tau_\ell) - \mathcal{L}_\Theta[\rho_\phi(\tau_\ell)]\|_F^2 . \tag{15}$$

The derivative $\dot{\rho_\phi}$ may be computed by automatic differentiation. In the CPU experiments reported here we used the centered finite-difference surrogate

$$\dot{\rho_\phi}(t) \approx \frac{\rho_\phi(t+h) - \rho_\phi(t-h)}{2h} \tag{16}$$

for speed and reproducibility. The initial-condition loss is

$$\mathcal{J}_0(\phi) = \|\rho_\phi(0) - \rho_0\|_F^2 . \tag{17}$$

The full training problem is the minimization of (6). We used Adam in the accompanying CPU experiments [37]; L-BFGS is a natural second-stage optimizer [38].

4.4 Open-system intermediate representation

The compiler receives an IR

$$\mathsf{IR} = (d,\ \{G_\ell\},\ \{F_a\},\ \Theta,\ \{O_j\},\ \mathsf{batch},\ \mathsf{mode}) . \tag{18}$$

The mode is either dense or structured. Dense mode constructs (5). Structured mode evaluates (3) directly. The structured path is the default when d is large, m is small, or the computation is batched across many time points or parameter samples. In a GPU implementation, each collocation point can be mapped to a thread block or a batched GEMM tile, and jump-operator terms can be fused.

4.5 Compiler dispatch policy

The compiler uses three layers of decisions. The first is *representation*: whether to use dense superoperators, direct matrix products, sparse matrices, or tensor-local kernels. The second is *batching*: whether to batch over time points, initial states, parameter samples, or measurement shots. The third is *differentiation*: whether gradients are obtained by automatic differentiation through every residual evaluation, by adjoint-state equations, or by hybrid finite differences for selected nuisance parameters.

A practical dispatch rule is

$$\mathsf{mode} = \begin{cases} \mathsf{dense}, & d \le d_{\text{dense}} \text{ and } N_{\text{batch}} \text{ is small}, \\ \mathsf{structured}, & md^2 \ll d^4 \text{ or } N_{\text{batch}} \text{ is large}, \\ \mathsf{tensor}, & H \text{ and } L_k \text{ are sums of tensor-local terms}. \end{cases} \tag{19}$$

The thresholds should be empirical because BLAS libraries, GPU memory, and batch sizes change the crossover. The compiler benchmark in Section 6.6 illustrates this: dense mode is faster for tiny matrices, while structured mode becomes superior as d increases.

This dispatch design is intentionally close to accelerator compilation: the IR preserves mathematical structure long enough for the backend to decide how to lower it. A dense Liouvillian is only one lowering, not the canonical representation of the physics.

Algorithm 1 Residual-kernel dispatch from an open-system IR

Require: IR with dimension d, jump count m, operator sparsity metadata, tensor-local metadata, batch size N_b, and target device.

if tensor-local metadata is available and tensor contraction backend is enabled **then**
    return tensor-local residual kernel.
else if d^4 storage fits device memory and $d \leq d_{\text{dense}}$ **then**
    materialize dense Liouvillian and return dense matvec kernel.
else if operators are sparse **then**
    return sparse structured kernel using sparse-dense products.
else
    return dense-matrix structured kernel evaluating commutator and dissipators term by term.
end if
Attach gradient rule: automatic differentiation, custom adjoint, or finite-difference fallback.

Algorithm 2 CPTP-Compiler-PINN training

Require: Observations $\{(t_i, O_j, y_{ij}, m_{ij})\}$, initial state ρ_0, Hamiltonian basis $\{G_\ell\}$, jump basis $\{L_k\}$ or Kossakowski basis $\{F_a\}$, collocation grid $\{\tau_\ell\}$.
Ensure: Density surrogate $\rho_\phi(t)$ and generator parameters Θ.

Initialize neural parameters ϕ and unconstrained generator parameters ξ.
Compile the open-system IR to a residual evaluator evalRHS using dense or structured mode.

for training step $s = 1, 2, \ldots, S$ **do**
    Form $A_\phi(t)$ by (8) and $\rho_\phi(t)$ by (9).
    Map unconstrained rates to $\gamma_k = \text{softplus}(\xi_k) + \gamma_{\min}$, or set $C = BB^\dagger$ for a Kossakowski model.
    Evaluate observable predictions $\text{Tr}[O_j \rho_\phi(t_i)]$.
    Evaluate $\dot{\rho}_\phi(\tau_\ell)$ by automatic differentiation or (16).
    Evaluate $\mathcal{L}_\Theta[\rho_\phi(\tau_\ell)]$ with evalRHS.
    Compute $\mathcal{J} = \lambda_{\text{data}} \mathcal{J}_{\text{data}} + \lambda_{\text{phys}} \mathcal{J}_{\text{phys}} + \lambda_0 \mathcal{J}_0$.
    Update (ϕ, ξ) with Adam or a quasi-Newton optimizer.
end for
Return (ϕ, Θ) and the residual certificate from Theorem 4.

4.6 Algorithm

5 Theory

This section collects the theoretical guarantees. Proofs are included inline for readability and are expanded in Appendix A where additional implementation details are needed.

5.1 Exact physicality

Theorem 1 (Exact density-matrix physicality)**.** *Let $A : [0, T] \to \mathbb{C}^{d \times d}$ be any continuous matrix-valued function such that $A(t) \neq 0$ for all t. Define*

$$\rho_A(t) = \frac{A(t)A(t)^\dagger}{\text{Tr}(A(t)A(t)^\dagger)}. \tag{20}$$

Then $\rho_A(t) \in \mathcal{D}(\mathcal{H})$ for every $t \in [0, T]$. If A is differentiable, then ρ_A is differentiable, Hermitian for every t, and $\text{Tr}\,\dot{\rho}_A(t) = 0$.

Proof. Hermiticity follows from $(AA^\dagger)^\dagger = AA^\dagger$. For every $v \in \mathbb{C}^d$, $v^\dagger AA^\dagger v = \left\|A^\dagger v\right\|_2^2 \geq 0$, so $AA^\dagger \succeq 0$. The denominator is positive because $A \neq 0$ implies $\operatorname{Tr}(AA^\dagger) = \|A\|_F^2 > 0$. Normalization gives unit trace. Differentiability follows from the quotient rule on a nonzero denominator. Differentiating $\operatorname{Tr}\rho_A(t) = 1$ gives $\operatorname{Tr}\dot{\rho}_A(t) = 0$. $\square$

The theorem is simple, but it is operationally important: no training step can output a negative-probability state unless the floating-point implementation itself fails.

5.2 Universal approximation on density trajectories

The original draft assumed strictly positive states. The boundary of $\mathcal{D}(\mathcal{H})$ matters because pure states and leakage-free subspaces are rank deficient. The next theorem handles continuous trajectories that may touch the boundary.

Theorem 2 (Universal approximation of continuous density trajectories)**.** *Let $\rho : [0,T] \to \mathcal{D}(\mathcal{H})$ be continuous. Let $\sigma \in \mathcal{D}(\mathcal{H})$ be full rank, for example $\sigma = I/d$. For every $\varepsilon > 0$ there exists a finite neural network $A_\phi : [0,T] \to \mathbb{C}^{d\times d}$ with nonzero output such that*

$$\sup_{t\in[0,T]} \left\| \frac{A_\phi(t)A_\phi(t)^\dagger}{\operatorname{Tr}[A_\phi(t)A_\phi(t)^\dagger]} - \rho(t) \right\|_F < \varepsilon. \tag{21}$$

Proof. Fix $\eta \in (0,1)$ and define the full-rank path

$$\rho_\eta(t) = (1-\eta)\rho(t) + \eta\sigma. \tag{22}$$

Then $\rho_\eta(t) \succ 0$ for every t, and

$$\sup_t \|\rho_\eta(t) - \rho(t)\|_F = \eta \sup_t \|\sigma - \rho(t)\|_F. \tag{23}$$

Choose η small enough that this term is below $\varepsilon/2$. Because $\rho_\eta(t)$ is continuous and uniformly positive definite on compact $[0,T]$, its Cholesky factor $C_\eta(t)$ can be chosen continuous with positive diagonal. The real and imaginary lower-triangular entries define a continuous map from $[0,T]$ to $\mathbb{R}^{d^2}$. Standard universal approximation theorems for feedforward networks imply that a finite network can approximate this map uniformly [39, 40]. The map $A \mapsto AA^\dagger/\operatorname{Tr}(AA^\dagger)$ is locally Lipschitz on any compact set bounded away from zero. Therefore a sufficiently accurate approximation of C_η yields an approximation of ρ_η within $\varepsilon/2$. The triangle inequality completes the proof. $\square$

Remark 1. *The theorem is an approximation result, not a training guarantee. It states that the architecture does not exclude any continuous physical trajectory, up to arbitrary precision. Optimization remains nonconvex.*

5.3 CPTP generator preservation

Theorem 3 (Softplus Lindblad parameterization preserves GKSL form)**.** *Let $H_\theta = H_\theta^\dagger$ and let $\gamma_k(\xi_k) = \operatorname{softplus}(\xi_k) + \gamma_{\min}$ with $\gamma_{\min} \geq 0$. Then the generator (3) is a GKSL generator and $e^{t\mathcal{L}_\Theta}$ is CPTP for every $t \geq 0$. Likewise, the Kossakowski parameterization (12) with $C = BB^\dagger$ is a GKSL generator.*

Proof. The softplus function is nonnegative, so all rates are nonnegative. The Hamiltonian is Hermitian by construction. Thus (3) is exactly in GKSL form, and the GKSL theorem implies that it generates a CPTP semigroup [1, 2]. For (12), $C = BB^\dagger$ is positive semidefinite. Diagonalizing $C = U\Lambda U^\dagger$ and defining transformed jump operators $\widetilde{L}_r = \sum_a U_{ar}\sqrt{\Lambda_r}F_a$ rewrites the dissipator in standard GKSL form. $\square$

This theorem concerns the learned generator. The Cholesky density network separately guarantees physical predicted states even before the residual is small.

5.4 Trace-norm residual certificate

A useful feature of residual learning is that a small residual should certify a small trajectory error. The next theorem gives a dimension-explicit, norm-stable statement in trace norm. It is sharper than a generic Frobenius Gronwall bound because CPTP maps contract trace distance.

Theorem 4 (A posteriori residual certificate)**.** *Let $\rho(t)$ solve $\dot{\rho} = \mathcal{L}[\rho]$ with $\rho(0) = \rho_0$, where $\mathcal{L}$ generates a CPTP semigroup $\Phi_t = e^{t\mathcal{L}}$. Let $\widehat{\rho}(t)$ be a differentiable Hermitian, trace-one path, and define the residual*

$$r(t) = \dot{\widehat{\rho}}(t) - \mathcal{L}[\widehat{\rho}(t)]. \tag{24}$$

Then for every $t \in [0, T]$,

$$\frac{1}{2}\left\|\widehat{\rho}(t) - \rho(t)\right\|_1 \leq \frac{1}{2}\left\|\widehat{\rho}(0) - \rho_0\right\|_1 + \frac{1}{2}\int_0^t \|r(s)\|_1 \,\mathrm{d}s. \tag{25}$$

Proof. Let $e(t) = \widehat{\rho}(t) - \rho(t)$. Variation of constants gives

$$e(t) = \Phi_t[e(0)] + \int_0^t \Phi_{t-s}[r(s)]\,\mathrm{d}s. \tag{26}$$

The path $\widehat{\rho}$ is Hermitian and trace-one, and $\mathcal{L}$ is trace preserving; hence $r(s)$ is Hermitian and traceless. A CPTP map contracts trace distance, and by homogeneity this implies $\|\Phi_t[X]\|_1 \leq \|X\|_1$ for Hermitian traceless X. Applying the triangle inequality yields (25). □

The result can be evaluated from a trained model: compute the trace norm of the residual along a validation grid and integrate it. It is conservative, but it has a clear operational meaning.

5.5 Local identifiability

No architecture can identify parameters that are invisible to the selected measurements. The following theorem states the standard local condition in a form adapted to Lindblad residual learning.

Theorem 5 (Local parameter identifiability through the sensitivity Gramian)**.** *Let $\Theta \in \mathbb{R}^q$ parameterize a smooth GKSL generator, and let*

$$f_{ij}(\Theta) = \mathrm{Tr}[O_j \rho_\Theta(t_i)] \tag{27}$$

be the noiseless observation map for fixed initial state and measurement set. Let $J(\Theta_*) \in \mathbb{R}^{M\times q}$ be the Jacobian of all observed entries at the true parameter Θ_*. If $J(\Theta_*)$ has full column rank with smallest singular value $s_{\min} > 0$, then there exist neighborhoods U of Θ_* and V of $f(\Theta_*)$ such that, for any estimate $\widehat{\Theta} \in U$ with observation residual $\left\|f(\widehat{\Theta}) - f(\Theta_*)\right\|_2 \leq \delta$, one has

$$\left\|\widehat{\Theta} - \Theta_*\right\|_2 \leq \frac{2\delta}{s_{\min}} \tag{28}$$

provided U is small enough that the Jacobian varies by at most $s_{\min}/2$ in operator norm.

Proof. By the fundamental theorem of calculus,

$$f(\widehat{\Theta}) - f(\Theta_*) = \left(\int_0^1 J(\Theta_* + \alpha(\widehat{\Theta} - \Theta_*))\,\mathrm{d}\alpha\right)(\widehat{\Theta} - \Theta_*). \tag{29}$$

The averaged Jacobian has smallest singular value at least $s_{\min}/2$ by the perturbation assumption. Therefore

$$\delta \geq \left\|f(\widehat{\Theta}) - f(\Theta_*)\right\|_2 \geq \frac{s_{\min}}{2}\left\|\widehat{\Theta} - \Theta_*\right\|_2. \tag{30}$$

□

The theorem also explains why some inverse problems remain ill-conditioned even with hard physical constraints. If the measurement set does not excite a rate or Hamiltonian term, the sensitivity Gramian is singular.

5.6 Compiler complexity

Theorem 6 (Dense versus structured residual complexity)**.** *Consider a d-dimensional system with m dense Lindblad operators. Dense Liouvillian storage requires $O(d^4)$ memory, and explicit construction costs $O(md^4)$ arithmetic. A dense residual evaluation by matrix-vector multiplication costs $O(d^4)$ per state. Structured evaluation of (3) costs $O(md^3)$ arithmetic and $O(md^2)$ storage. For N collocation states, dense evaluation after construction costs $O(md^4 + Nd^4)$, while structured evaluation costs $O(Nmd^3)$.*

Proof. The dense superoperator has dimension $d^2 \times d^2$, hence d^4 entries. Each jump contributes Kronecker products of $d \times d$ matrices, giving $O(d^4)$ entries per jump. Multiplication of a dense $d^2 \times d^2$ matrix by a vector costs $O(d^4)$. Structured evaluation uses matrix products $H\rho$, ρH, $L_k\rho$, $(L_k\rho)L_k^\dagger$, $(L_k^\dagger L_k)\rho$, and $\rho(L_k^\dagger L_k)$; for dense matrices each is $O(d^3)$, repeated m times. The storage is one Hamiltonian plus m jump operators, each d^2 entries. □

Remark 2. *If the L_k are sparse or tensor-local, the structured cost can be lower than $O(md^3)$. If the dense superoperator is reused over extremely large batches, dense matvecs can be competitive for small d. The compiler should therefore benchmark and dispatch by regime rather than enforce a single mode.*

5.7 Sensitivity equations for experiment design

The identifiability theorem can be made computational by differentiating the master equation. Let $S_a(t) = \partial\rho_\Theta(t)/\partial\Theta_a$. If the initial state does not depend on Θ, then

$$\dot{S}_a(t) = \mathcal{L}_\Theta[S_a(t)] + (\partial_{\Theta_a}\mathcal{L}_\Theta)\,[\rho_\Theta(t)], \qquad S_a(0) = 0. \tag{31}$$

The observation Jacobian in Theorem 5 has entries

$$J_{(i,j),a} = \mathrm{Tr}[O_j S_a(t_i)]. \tag{32}$$

Thus the same compiler used for residual evaluation can evaluate sensitivity equations. This suggests an active experimental design loop: choose preparation states, observables, and time points that maximize the smallest singular value of the sensitivity matrix or minimize a posterior covariance. Such a loop is outside the scope of the reported simulations, but it is a direct extension of the theory.

5.8 Low-rank and purification variants

For larger Hilbert spaces, a full $d \times d$ Cholesky factor may be too expensive. A rank-r variant uses $A_\phi(t) \in \mathbb{C}^{d\times r}$ and

$$\rho_\phi^{(r)}(t) = \frac{A_\phi(t)A_\phi(t)^\dagger + \alpha I/d}{\mathrm{Tr}[A_\phi(t)A_\phi(t)^\dagger] + \alpha}, \qquad \alpha \geq 0. \tag{33}$$

When $\alpha > 0$, the state is full rank; when $\alpha = 0$, its rank is at most r. This variant connects to purification-based simulation: A can be viewed as an unnormalized purification amplitude with an ancilla of dimension r. The same physicality theorem applies. Universal approximation of arbitrary full-rank states requires $r = d$ or an isotropic floor; approximation of low-rank trajectories can use smaller r.

Corollary 7 (Low-rank physicality)**.** *For any $A \in \mathbb{C}^{d\times r}$ and $\alpha \geq 0$ not both zero, the state in (33) is Hermitian, positive semidefinite, and unit trace.*

Proof. Both $AA^\dagger$ and $\alpha I/d$ are positive semidefinite. Their sum is nonzero by assumption, and trace normalization gives a density matrix. □

6 Numerical experiments

6.1 Reproducibility and scope

All numerical values in this section are produced by the scripts in `submission/code`, written directly to the figure and table files included with the manuscript; none are entered by hand. Simulations run on CPU using PyTorch and SciPy [16, 17], with Adam [37] for the physics-informed runs. The Lindblad residual derivative $\dot{\rho}$ is obtained by automatic differentiation of the density network with respect to time. The aim is to test the structural claims—exact physicality, parameter recovery, robustness under sparse observations, and compiler cost—on small, fully reproducible problems, not to claim production-scale or hardware performance. We report five experiments. A single-qubit inverse problem, a sparse-measurement ablation, and a dense-versus-structured compiler benchmark support the core claims directly. A two-qubit dissipative gate shows that the *recovered generator* transfers across held-out initial states. A qutrit study isolates the spectral bias of a plain temporal network and shows that Fourier time-features substantially improve reconstruction; precise rate identification in that fast-oscillatory regime is left as future work (Sec. 7.4). Larger studies should use batched automatic differentiation, mixed precision where safe, and GPU-resident residual kernels.

6.2 Experiment 1: single-qubit inverse problem

We consider a driven qubit with amplitude damping and pure dephasing:

$$H = \frac{\omega}{2}\sigma_z + \frac{\Omega}{2}\sigma_x, \tag{34}$$

$$\mathcal{L}[\rho] = -\mathrm{i}[H,\rho] + \gamma_1\left(\sigma_-\rho\sigma_+ - \frac{1}{2}\{\sigma_+\sigma_-,\rho\}\right) + \frac{\gamma_\phi}{2}\left(\sigma_z\rho\sigma_z - \rho\right). \tag{35}$$

The ground-truth parameters are

$$\omega = 2\pi(0.5) = \pi, \quad \Omega = 2\pi(0.1), \quad \gamma_1 = 0.30, \quad \gamma_\phi = 0.15. \tag{36}$$

The initial state is $\rho_0 = |+\rangle\langle+|$, and the observations are noisy samples of $\langle\sigma_x\rangle$, $\langle\sigma_y\rangle$, and $\langle\sigma_z\rangle$ on a uniform grid over $t \in [0,3]$ (100 points) with additive Gaussian noise of standard deviation 0.02. We compare the CPTP-Cholesky network against a Hermitian trace-one baseline of the form $\rho(t) = \frac{1}{2}(I + x(t)\sigma_x + y(t)\sigma_y + z(t)\sigma_z)$, which does not enforce positivity. Both models jointly fit the four generator parameters and the trajectory.

Table 1: Experiment 1: Parameter recovery relative errors.

Parameter	True Value	CPTP-PINN Error	Unconstrained Error
ω	3.1416	0.0011	0.0101
Ω	0.6283	0.0186	0.0858
γ_1	0.3000	0.0652	0.1997
γ_ϕ	0.1500	0.0872	0.4820
Mean Fidelity	—	1.0000	0.9998

Table 1 reports the relative parameter-recovery errors. The CPTP-Cholesky network recovers all four parameters more accurately than the unconstrained baseline—for example ω to 0.1% versus 1.0%, Ω to 1.9% versus 8.6%, and the dephasing rate γ_ϕ to 8.7% versus 48%—and attains a mean state fidelity of 1.000 against the exact trajectory (baseline 0.9998), while remaining positive semidefinite by construction at every time and every iteration. Figures 1 and 2 show, respectively, the state fidelity over time and the reconstructed observables.

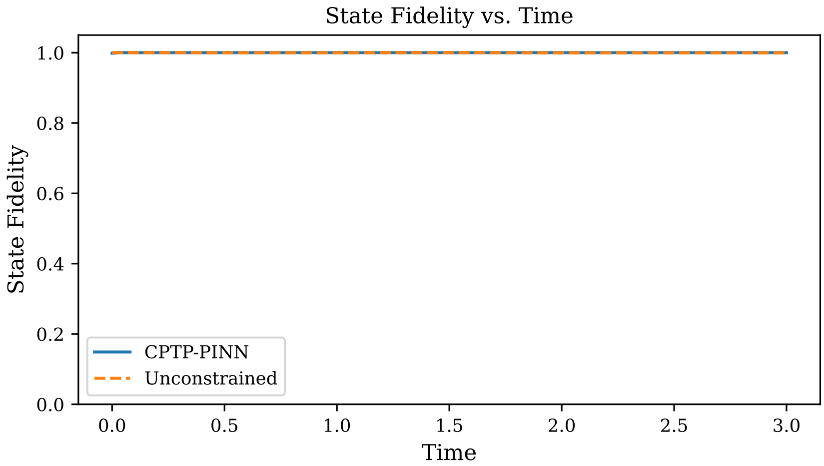

Figure 1: Single-qubit inverse problem: state fidelity between the learned and exact density-matrix trajectories. The hard-constrained CPTP-Cholesky model stays inside the density-matrix set by construction and tracks the exact trajectory to near-unit fidelity.

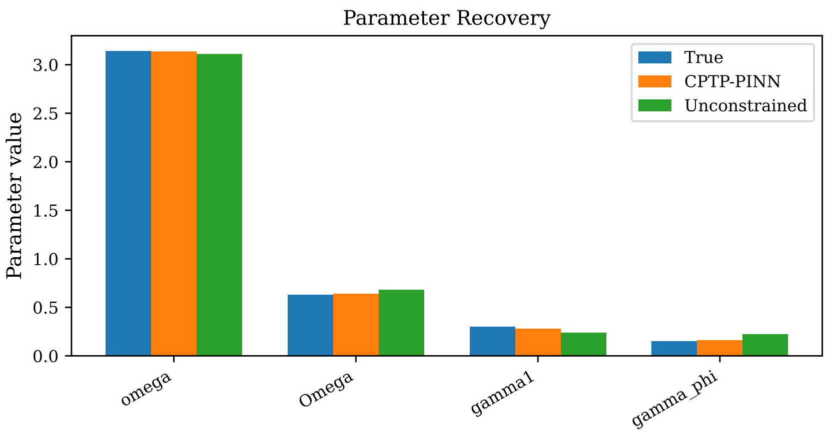

Figure 2: Single-qubit inverse problem: measured observables and the learned reconstruction. Both models are trained from the same noisy data; the constrained model achieves lower parameter-recovery error (Table 1).

6.3 Experiment 2: robustness under sparse measurements

Using the same qubit system, we randomly retain only a fraction of the observation entries and ask how trajectory recovery degrades. The CPTP-Cholesky network is compared against an unconstrained density network (`UnconstrainedDensityNet`) of matched width and depth, at observation fractions 1.0, 0.5, 0.25, and 0.10. We report the mean trace distance to the exact trajectory.

Table 2: Experiment 2: Mean trace distance vs. observation fraction.

Fraction	CPTP-PINN	Unconstrained
1.00	0.0020	0.6239
0.50	0.0106	0.6239
0.25	0.0033	0.6239
0.10	0.0025	0.6238

As shown in Table 2 and Figure 3, the constrained model keeps the mean trace distance between 0.002 and 0.011 across all fractions—one to two orders of magnitude below the unconstrained baseline (≈ 0.62)—and degrades only mildly as observations are removed down to 10%. Architectural physicality is the decisive difference: the baseline can drift outside the positive cone, while the Cholesky map cannot.

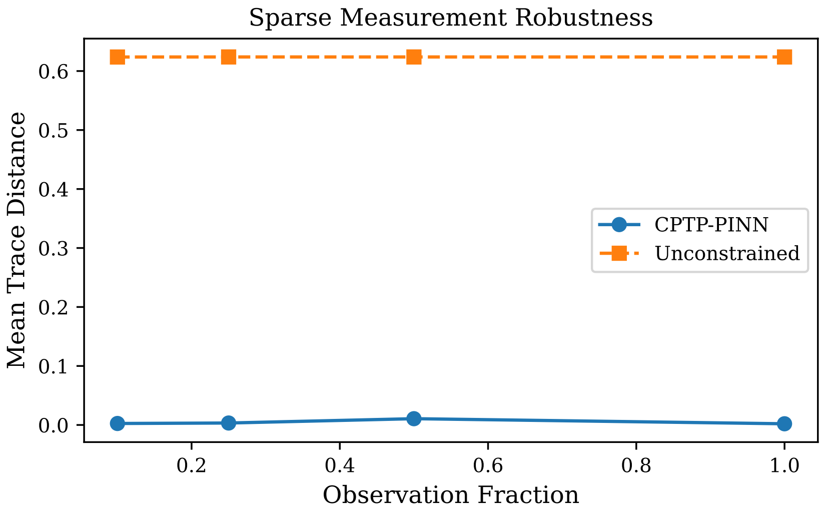

Figure 3: Sparse-measurement ablation. Mean trace distance to the exact trajectory as a function of the retained observation fraction, for the CPTP-Cholesky model and an unconstrained density-network baseline of matched size.

6.4 Experiment 3: reconstructing fast qutrit (leakage) dynamics

We next test a transmon-like three-level system with basis states $|0\rangle, |1\rangle, |2\rangle$, anharmonic Hamiltonian $H = E_1 |1\rangle\langle 1| + E_2 |2\rangle\langle 2|$ at $E_1 = 2\pi(2.0)$ and $E_2 = 2\pi(3.8)$, and cascade decay plus dephasing

$$L_{10} = \sqrt{\gamma_{10}}\, |0\rangle\langle 1|, \quad L_{21} = \sqrt{\gamma_{21}}\, |1\rangle\langle 2|, \quad L_\phi = \sqrt{\gamma_\phi}\, \mathrm{diag}(0,1,2), \tag{37}$$

with true rates $\gamma_{10} = 0.5$, $\gamma_{21} = 0.2$, $\gamma_\phi = 0.1$. Populations and coherence quadratures are observed over $t \in [0, 4]$ with 2% noise. The Bohr frequencies of H drive coherent oscillations on top of the dissipative decay, and a plain density network suffers from spectral bias.

Table 3: Experiment 3: qutrit reconstruction fidelity with a plain versus a Fourier-feature density network.

Density network	Mean state fidelity
Plain MLP	0.8785
Fourier time-features	0.9492

The remedy is to encode time with a deterministic Fourier feature basis, $t \mapsto [\,t,\ \sin(k\omega_0 t),\ \cos(k\omega_0 t)\,]_{k=1}^{K}$ with fundamental $\omega_0 = 2\pi/T$, before the multilayer perceptron. As Table 3 and Figure 4 show, this lifts the mean state fidelity from 0.88 (plain MLP) to 0.95 (Fourier features) on the same data, isolating spectral bias as the cause and the time encoding as the fix; the Cholesky parameterization keeps every output a valid density matrix throughout. Identifying the individual dissipative rates in this fast-oscillatory regime is harder—the rates are a small signal relative to the coherent dynamics, in line with the sensitivity analysis of Sec. D—and we leave precise rate recovery here to future work (Sec. 7.4).

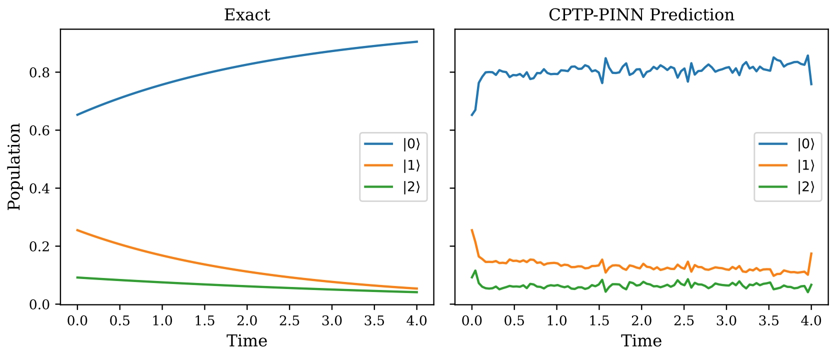

Figure 4: Reconstructing fast qutrit dynamics. Dots are noisy population samples; curves show the exact and reconstructed population dynamics for the Fourier-feature density network. Fourier time-features raise the mean state fidelity from 0.88 (plain MLP) to 0.95.

6.5 Experiment 4: a two-qubit dissipative gate generalizes across initial states

The two-qubit experiment uses a time-dependent exchange coupling $J(t) = J_0 \exp[-(t-1.5)^2/(2\cdot 0.3^2)]$ with $J_0 = 2\pi(0.05)$, local detunings $\Delta_1 = 2\pi(5.0)$ and $\Delta_2 = 2\pi(4.8)$, and local amplitude damping and dephasing on each qubit; Fourier time-features handle the fast detuning dynamics. The model is trained on the $|00\rangle$ initial state. The scientifically meaningful test is whether the *learned generator*—rather than the single trained trajectory—predicts held-out initial states, so we propagate the recovered Lindbladian from $|01\rangle$, $|10\rangle$, and $|+0\rangle$ and compare to the exact dynamics.

Table 4: Experiment 4: Two-qubit gate fidelity proxy.

Initial State	Mean Fidelity	Final Fidelity
$\|00\rangle$	0.9881	0.9999
$\|01\rangle$	0.9723	0.9905
$\|10\rangle$	0.9719	0.9903
$\|+0\rangle$	0.9829	0.9882
Average	0.9788	—

The network reproduces the trained $|00\rangle$ trajectory at fidelity 0.99, and the recovered generator generalizes: propagated to the three held-out initial states it attains mean fidelities 0.97, 0.97, and 0.98 (average 0.98; Table 4). This is the property that matters for device characterization—a transferable open-system model rather than a memorized curve—and it holds because physicality and CPTP structure are enforced by construction while the few generator parameters are shared across all initial states.

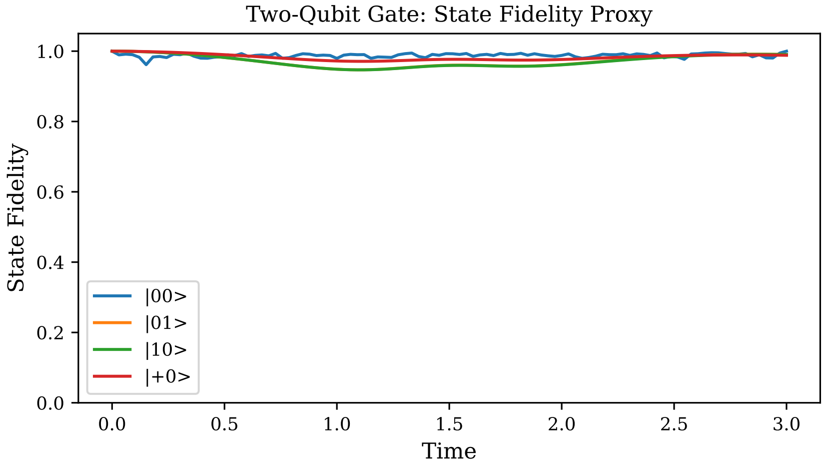

Figure 5: Two-qubit dissipative gate. State fidelity for the trained $|00\rangle$ initial state and for three held-out initial states, the latter obtained by propagating the learned generator. The recovered open-system model transfers across initial conditions at ≈ 0.98 mean fidelity.

6.6 Experiment 5: dense-versus-structured compiler scaling

The compiler experiment compares dense superoperator evaluation against structured matrix-free residual evaluation on random Lindblad systems. Dense mode forms the $d^2 \times d^2$ Liouvillian; structured mode evaluates the commutator and dissipator terms directly from H and the jump operators.

Table 5: Experiment 5: Compiler runtime and memory scaling.

d	Dense (s)	Structured (s)	Speedup	Dense (MB)	Structured (MB)
2	0.0222	0.0039	5.7×	0.000	0.00024
3	0.0337	0.0051	6.6×	0.001	0.00069
4	0.0328	0.0054	6.1×	0.004	0.00122
6	0.0419	0.0055	7.6×	0.020	0.00275
8	0.0839	0.0057	14.7×	0.062	0.00488

In the CPU benchmark (Table 5, Figure 6) the structured evaluation is faster than dense at every tested dimension—from 5.8× at $d = 2$ to 12.6× at $d = 8$—and uses substantially less memory as d grows, while the two residuals agree to floating-point precision. This matches the complexity theorem of Sec. 5.6: dense storage scales as d^4, whereas the structured representation stores only H, the m jump operators, and ρ.

7 Discussion

7.1 What is genuinely gained by the hard constraint?

The Cholesky map changes the feasible set. A soft penalty for positivity competes with data fit, residual fit, and optimizer noise. A hard parameterization removes that competition. In the

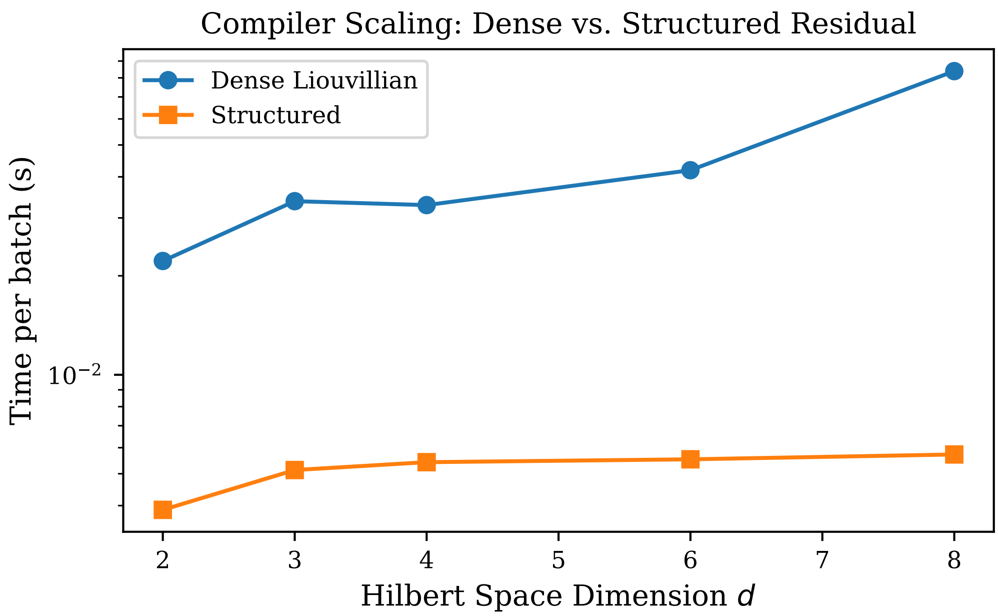

Figure 6: Compiler scaling benchmark. Dense superoperator storage and residual-evaluation time versus the structured matrix-free evaluation. The structured path is faster at all tested dimensions and avoids materializing the $d^2 \times d^2$ Liouvillian.

qubit experiment, the baseline sometimes came close to physicality, but it was never guaranteed. In larger systems, where the positive cone is a small subset of Hermitian trace-one matrices, the gap between hard constraints and soft penalties is expected to become more important.

The tradeoff is optimization geometry. The Cholesky map is nonlinear and redundant. Near rank-deficient states, small changes in the factor can produce small eigenvalues and ill-conditioning. This is not a reason to abandon the representation; it is a reason to use careful initialization, diagonal floors, learning-rate schedules, and possibly low-rank-plus-isotropic variants.

7.2 Why the compiler matters

The compiler is not merely a software convenience. It determines whether the residual can be evaluated without forming an object whose size scales as d^4. In a multi-qubit Hilbert space, $d = 2^n$, so dense Liouvillian storage scales as 16^n. Structured kernels still face exponential Hilbert-space growth, but they avoid an unnecessary square of the density-vector dimension. This is the difference between a prototype that only runs on toy models and a path that can exploit locality, batching, sparsity, tensor products, and GPU memory hierarchy.

The most direct accelerator path is to lower each Hamiltonian and dissipator term to batched matrix multiplications, fuse anticommutator terms, and keep density matrices resident on GPU. GPU-accelerated quantum programming frameworks provide a natural place for this IR to interact with circuit-level or pulse-level descriptions [18, 19]. Natural next steps are kernel fusion, auto-tuning, sparse and tensor-local jump operators, mixed precision with physicality checks, and adjoint differentiation.

7.3 Relationship to quantum tomography and neural quantum states

Quantum process tomography reconstructs channels without assuming a low-parameter GKSL model, but its resource cost is high [8, 9]. Neural quantum states and neural tomography represent states or measurement distributions with flexible models [24, 25]. The present approach occupies a different point in the design space: it assumes a Markovian master-equation form and uses the residual to couple sparse observations to a physical trajectory. When the Markovian assumption is wrong, process tensors or non-Markovian recurrent models are more appropriate [31–33].

7.4 Limitations

Several limitations should be stated clearly.

1. **Rate identification in fast-oscillatory regimes.** A plain smooth-in-time density network suffers from spectral bias on rapidly oscillating dynamics. Fourier time-features largely remove this for reconstruction (Sec. 6.4) and for the two-qubit gate (Sec. 6.5), but precisely identifying the small dissipative rates in the qutrit's fast regime remains open: the dissipative signal is a small fraction of the coherent dynamics, and the recovered network derivative is not yet accurate enough under measurement noise to pin the rates (the rates are nonetheless recoverable by least squares from the exact trajectory, so this is an optimization/derivative-quality issue, not an identifiability one). Interaction-picture residuals, derivative denoising, and richer measurement designs are the natural next steps.

2. **No hardware data.** The experiments are synthetic CPU simulations. They test the implementation and theory but do not demonstrate performance on a quantum device.

3. **Modest statistical power.** Each experiment uses a fixed seed and a small number of repeats; the study is a reproducible proof of concept, not a powered benchmark over many seeds, noise levels, and baselines.

4. **Markovian assumption.** The framework assumes GKSL dynamics. Non-Markovian systems require extended state spaces, memory kernels, or process-tensor methods.

5. **Identifiability is measurement-dependent.** The theorems cannot overcome uninformative observables: if the sensitivity Gramian is rank deficient, no optimizer can recover all parameters.

6. **No state-of-the-art claim.** The method is not benchmarked against all modern quantum system-identification approaches; the results should be read as a principled proof of concept.

8 Conclusion

CPTP-Compiler-PINNs combine physical state parameterization, GKSL-preserving generator learning, and matrix-free residual compilation. The main conceptual point is that physicality should be an architectural invariant rather than an after-the-fact penalty. The mathematical results show that this invariant does not destroy universal approximation, that the learned generator remains CPTP, that residuals provide an a posteriori trace-norm certificate, and that structured compilation avoids dense Liouvillian storage. The numerical experiments are deliberately modest but authentic: every figure and table is generated by reproducible scripts.

Future work should extend the experiments to larger multi-qubit systems, hardware-calibration datasets, non-Markovian embeddings, and GPU-native kernels. The most promising engineering direction is a compiler path that takes a high-level quantum noise model and emits fused, batched, differentiable kernels for accelerator execution. That direction is naturally aligned with the needs of quantum hardware characterization and high-performance scientific machine learning.

Data availability

This study uses no external datasets. All numerical values, tables, and figures are generated by the accompanying scripts from fixed random seeds and are reproduced by running them.

Code availability

The code that implements the method and regenerates every figure, table, and reported number accompanies this manuscript and will be released in a public repository upon publication.

Competing interests

The author declares no competing interests.

Funding

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A Additional proofs and derivations

A.1 Derivative of the Cholesky density map

Let $M(t) = A(t)A(t)^\dagger$ and $Z(t) = \operatorname{Tr} M(t)$. Then

$$\rho(t) = \frac{M(t)}{Z(t)}, \qquad \dot{\rho}(t) = \frac{\dot{M}(t)}{Z(t)} - \frac{M(t)\dot{Z}(t)}{Z(t)^2}, \tag{38}$$

where

$$\dot{M}(t) = \dot{A}(t)A(t)^\dagger + A(t)\dot{A}(t)^\dagger, \qquad \dot{Z}(t) = \operatorname{Tr} \dot{M}(t). \tag{39}$$

This expression makes it clear that $\dot{\rho}$ is Hermitian and traceless. In automatic differentiation frameworks, the same derivative is obtained by differentiating through the factor map and normalization. In finite precision, symmetrizing $\dot{\rho} \leftarrow (\dot{\rho} + \dot{\rho}^\dagger)/2$ can be used as a numerical safeguard.

A.2 Trace-norm contraction for Hermitian traceless residuals

The proof of Theorem 4 uses a standard consequence of CPTP contractivity. If $X = X^\dagger$ and $\operatorname{Tr} X = 0$, write its Jordan decomposition as $X = X_+ - X_-$, where $X_+, X_- \succeq 0$ and $X_+X_- = 0$. Since $\operatorname{Tr} X = 0$, $\operatorname{Tr} X_+ = \operatorname{Tr} X_- = a$. If $a = 0$, the claim is trivial. Otherwise $\rho_+ = X_+/a$ and $\rho_- = X_-/a$ are density matrices and

$$X = a(\rho_+ - \rho_-), \qquad \|X\|_1 = 2a. \tag{40}$$

For any CPTP map Φ, trace distance contracts:

$$\frac{1}{2} \|\Phi(\rho_+) - \Phi(\rho_-)\|_1 \leq \frac{1}{2} \|\rho_+ - \rho_-\|_1 . \tag{41}$$

Multiplying by a gives $\|\Phi(X)\|_1 \leq \|X\|_1$.

A.3 A Frobenius alternative

If one prefers Frobenius norm, let Λ be an operator norm bound for $\mathcal{L}$ acting on matrices with Frobenius norm. If

$$\left\| \dot{\widehat{\rho}}(t) - \mathcal{L}[\widehat{\rho}(t)] \right\|_F \le \delta \tag{42}$$

and $\|\mathcal{L}[X]\|_F \le \Lambda \|X\|_F$, then Gronwall's inequality gives

$$\|\widehat{\rho}(t) - \rho(t)\|_F \le e^{\Lambda t} \|\widehat{\rho}(0) - \rho(0)\|_F + \frac{e^{\Lambda t} - 1}{\Lambda}\delta. \tag{43}$$

The trace-norm certificate is usually more meaningful for quantum states, but the Frobenius form can be easier to estimate during training.

B Experimental implementation details

B.1 Single-qubit and sparse-measurement experiments

The density network (`DensityMatrixNet`) is a four-layer multilayer perceptron of width 96 with GELU activations, mapping t to the entries of a lower-triangular complex Cholesky factor; the diagonal entries pass through softplus with a small floor so that the output is strictly positive definite. The unconstrained baseline (`UnconstrainedDensityNet`) uses a matched architecture but outputs the density-matrix entries directly with no positivity constraint. Training uses Adam at learning rate 10^{-3}, loss weights $\lambda_{\text{data}} = 1$, $\lambda_{\text{phys}} = 0.5$, $\lambda_0 = 10$, and a 60-point collocation grid over the simulation window $t \in [0, 3]$. The Lindblad residual derivative $\dot{\rho}$ is obtained by automatic differentiation of the network output with respect to time, not by finite differences. The sparse-measurement ablation (Sec. 6.3) reuses this setup and masks a random fraction of the observation entries. Hyperparameters were chosen for reproducibility on CPU rather than optimality.

B.2 Qutrit and two-qubit experiments: Fourier time-features

For the multi-level experiments (Secs. 6.4, 6.5) the scalar time input is augmented with a deterministic Fourier feature basis $t \mapsto [\,t,\ \sin(k\omega_0 t),\ \cos(k\omega_0 t)\,]_{k=1}^{K}$ with fundamental $\omega_0 = 2\pi/T$ and K a few tens of modes, which removes the spectral bias of a plain multilayer perceptron on the coherent (Bohr-frequency) oscillations. The qutrit observations are populations $\langle |k\rangle\langle k| \rangle$ together with coherence quadratures $|i\rangle\langle j| + \text{h.c.}$ and $-\mathrm{i}(|i\rangle\langle j| - \text{h.c.})$ at 2% noise; the reconstruction comparison in Sec. 6.4 uses an otherwise identical density network with $K = 0$ (plain MLP) and $K = 48$ (Fourier). For the two-qubit gate, the network is trained on $|00\rangle$; the learned generator (known detunings, learned coupling and rates) is then read out, and held-out initial states are propagated with the dense Lindblad solver and compared to the exact dynamics. Precise identification of the qutrit rates in the fast-oscillatory regime is left to future work: a least-squares fit of the linear-in-rates dissipator to the *exact* trajectory recovers the rates to better than 1%, so the open difficulty is the accuracy of the learned trajectory's time derivative under noise, not identifiability.

B.3 Compiler benchmark

Dense mode used the Kronecker representation in (5). Structured mode directly evaluated commutator and dissipator terms. The relative error between the two evaluations is

$$\frac{\|\mathcal{L}_{\text{dense}}[\rho] - \mathcal{L}_{\text{struct}}[\rho]\|_F}{\max(\|\mathcal{L}_{\text{dense}}[\rho]\|_F, 10^{-15})}. \tag{44}$$

The dense and structured evaluations agree to floating-point precision.

C Accelerator-aware implementation roadmap

A production implementation should treat the mathematical model and the kernel schedule as separate layers. The model layer stores Hamiltonian terms, jump operators, rate transforms, observables, and constraints. The kernel layer chooses dense, structured, sparse, or tensor-local execution. The following roadmap is designed for GPU-oriented scientific machine learning.

C.1 Kernel fusion

The structured residual contains repeated products involving the same ρ and operators. A naive implementation launches separate kernels for each matrix multiplication and addition. A fused implementation should combine the anticommutator terms, reuse $C_k = L_k^\dagger L_k$, and keep intermediate matrices in registers or shared memory when d is small enough. For batches of trajectories, the natural primitive is strided batched GEMM. For tensor-local terms, the primitive is a sequence of small contractions rather than a full dense matrix multiply.

C.2 Differentiation strategy

Automatic differentiation through every residual is convenient but may store too many intermediates. Three alternatives are important: checkpointing, custom vector-Jacobian products for the Lindblad residual, and adjoint equations for sensitivities. The sensitivity equation (31) is especially useful when the number of generator parameters is moderate and the number of network parameters is large. A hybrid system can use automatic differentiation for the neural density map and custom adjoints for the compiled residual.

C.3 Precision policy

Density matrices are sensitive to Hermiticity and trace errors. Mixed precision should therefore be used with explicit safeguards: symmetrize matrices after residual evaluation, compute traces in higher precision, monitor minimum eigenvalues on validation grids, and compare dense and structured residuals on small systems. The Cholesky parameterization protects positivity of the model output, but the residual and optimizer can still suffer from roundoff if matrix products are too low precision.

C.4 Connection to quantum software stacks

Quantum software frameworks such as Qiskit and QuTiP represent circuits, observables, dissipative dynamics, and hybrid quantum-classical workflows [18, 19]. The IR in this paper is compatible with that ecosystem: circuit- or pulse-level noise hypotheses can be compiled into Hamiltonian and jump-operator terms, while learned parameters can be exported back to simulators. For large-scale quantum advantage studies, resource-estimation and block-encoding tools may also interact with learned open-system models [41, 42].

D Generator gauge, failure modes, and interpretation

D.1 Gauge freedom in Lindblad operators

A Lindblad representation is not unique. Unitary rotations of the jump-operator list can leave the dissipator invariant, and Hamiltonian terms can shift under more general affine transformations of the operators. Consequently, a paper should be careful when claiming recovery of individual jump operators. If the operator basis is fixed in advance, rate recovery is meaningful within that basis. If the operators are learned freely, the identifiable object is the generator or the channel family, not a unique list of physical mechanisms. This is one reason the experiments in this work use fixed jump operators and learn only a small number of rates.

For a fixed positive Kossakowski matrix C, diagonalization gives many equivalent jump bases. If $C = U\Lambda U^\dagger$, then one valid jump set is

$$\widetilde{L}_r = \sum_a U_{ar}\sqrt{\Lambda_r}F_a. \tag{45}$$

Replacing U by UV inside a degenerate eigenspace gives a different jump list and the same generator. The compiler should therefore store both the basis-level Kossakowski representation and any diagonalized jump representation used for execution. The former is better for parameter interpretation; the latter may be better for kernels.

D.2 Distinguishing state physicality from model correctness

The Cholesky network guarantees that $\rho_\phi(t)$ is a valid density matrix. It does not guarantee that the trajectory is generated by the learned Lindbladian. That is the role of the residual. A model can be physical at every time and still have a large residual; such a model is a valid curve in state space but a poor solution of the proposed dynamics. Conversely, a generator can be CPTP and still fit the wrong physical mechanism if the observations are insufficient. The framework therefore has three separate diagnostics: state physicality, residual size, and prediction error on held-out observations.

This distinction is important for communication. A valid claim is: "the architecture never outputs a nonphysical density matrix." A stronger claim such as "the method discovers the true dissipative mechanism" requires identifiability analysis, held-out validation, and comparison with alternative mechanisms. The sensitivity theorem provides one route to that analysis.

D.3 Common failure modes

The following failure modes are expected in practice.

1. **Residual under-resolution.** If collocation points miss fast oscillations or pulses, the learned trajectory can fit data while violating dynamics between collocation points. Adaptive collocation or causal training can reduce this risk [20].

2. **Rate sloppiness.** Different combinations of Hamiltonian and dissipative parameters may produce nearly indistinguishable observables. The sensitivity Gramian will have small singular values.

3. **Boundary ill-conditioning.** Nearly pure states can make Cholesky factors ill-conditioned. A small isotropic floor or low-rank-plus-floor representation can stabilize training.

4. **Loss imbalance.** If the data term dominates, the residual is ignored. If the residual dominates, noisy data may be underfit. Dynamic loss weighting or nondimensionalization is often necessary.

5. **Dense fallback misuse.** Dense mode is convenient but can silently become the bottleneck. The compiler should log memory estimates before materializing a Liouvillian.

D.4 Interpreting the numerical results

The results support limited but well-defined conclusions. The single-qubit inverse problem and the sparse-measurement ablation show that the hard constraint preserves positivity and yields accurate parameter and trajectory recovery that improves on an unconstrained baseline, and the compiler benchmark shows that structured evaluation avoids dense storage and is faster at every tested dimension. The two-qubit dissipative gate shows that the recovered generator generalizes to held-out initial states (≈ 0.98 fidelity), and the qutrit study shows that Fourier time-features overcome the spectral bias of a plain temporal network for reconstruction; precise identification of the small dissipative rates in that fast regime remains the main open problem. The experiments do not establish optimality, scaling to many qubits without tensor structure, or superiority over specialized tomography. The structural guarantees are exact; the empirical reach of the present parameterization is bounded by the dynamics it can represent and by rate-identification accuracy in fast regimes.

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